

CYCLONES, STORM SURGE AND COASTAL PREPAREDNESS

COMPOUND CYCLONE HAZARD MAP → CYCLOGENESIS BOUNDARY → COASTAL WATER FIREWALL → IMD INFORMATION RAIL → COLOUR-STAGE MATRIX → WARNING-TO-EVACUATION CHAIN

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 COMPOUND CYCLONE HAZARD MAP

COMPOUND CYCLONE HAZARD MAP STRUCTURE DEBRIS POWER TELECOM RIVERINE FLASH PLUVIAL URBAN FLOOD STORM SURGE + WAVES COASTAL INUNDATION EROSION TRANSPORT HEALTH WATER AND LIVELIHOODS

WIND → structure debris power telecom

RAIN → riverine flash pluvial urban flood

STORM SURGE + WAVES → coastal inundation erosion

CASCADE → transport health water and livelihoods

HAZARD / RISK / RESILIENCE CONCEPT

- WIND → structure debris power telecom
- RAIN → riverine flash pluvial urban flood

INSTITUTIONAL OR COMMUNITY RESPONSE

- STORM SURGE + WAVES → coastal inundation erosion
- CASCADE → transport health water and livelihoods

CAPACITY / STATUS / EVIDENCE LIMIT

- WIND → structure debris power telecom
- RAIN → riverine flash pluvial urban flood

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STORM SURGE + WAVES → coastal inundation erosion.

01

STAGE 01 CYCLOGENESIS BOUNDARY

CYCLOGENESIS BOUNDARY WARM WATER + INSTABILITY + MOISTURE CORIOLIS + DISTURBANCE + LIMITED SHEAR CYCLONIC SYSTEM DM FOCUS EXPOSURE WARNING ACTION CONTINUITY

WARM WATER + INSTABILITY + MOISTURE

CORIOLIS + DISTURBANCE + LIMITED SHEAR

CYCLONIC SYSTEM

DM FOCUS → exposure warning action continuity

HAZARD / ACTOR / INSTRUMENT

- WARM WATER + INSTABILITY + MOISTURE
- CORIOLIS + DISTURBANCE + LIMITED SHEAR

DECISIVE RISK OR MANDATE DISTINCTION

- CYCLONIC SYSTEM
- DM FOCUS → exposure warning action continuity

WARNING / IMPLEMENTATION / OUTCOME LIMIT

- WARM WATER + INSTABILITY + MOISTURE
- CORIOLIS + DISTURBANCE + LIMITED SHEAR

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CYCLONIC SYSTEM; DM FOCUS → exposure warning action continuity.

02

STAGE 02 COASTAL WATER FIREWALL

COASTAL WATER FIREWALL STORM SURGE CYCLONE WIND AND PRESSURE ASTRONOMICAL GRAVITATIONAL CYCLE SUDDEN WATER DISPLACEMENT TOTAL WATER LEVEL INTERACTING BUT DISTINCT COMPONENTS

HAZARD / ACTOR / INSTRUMENT

STORM SURGE → cyclone wind and pressure
TIDE → astronomical gravitational cycle

DECISIVE RISK OR MANDATE DISTINCTION

TSUNAMI → sudden water displacement
TOTAL WATER LEVEL → interacting but distinct components

WARNING / IMPLEMENTATION / OUTCOME LIMIT

STORM SURGE → cyclone wind and pressure
TIDE → astronomical gravitational cycle

HAZARD / ACTOR / INSTRUMENT

- STORM SURGE → cyclone wind and pressure
- TIDE → astronomical gravitational cycle

DECISIVE RISK OR MANDATE DISTINCTION

- TSUNAMI → sudden water displacement
- TOTAL WATER LEVEL → interacting but distinct components

WARNING / IMPLEMENTATION / OUTCOME LIMIT

- STORM SURGE → cyclone wind and pressure
- TIDE → astronomical gravitational cycle

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TOTAL WATER LEVEL → interacting but distinct components.

03

STAGE 03 IMD INFORMATION RAIL

IMD INFORMATION RAIL 1 OBSERVE 2 CLASSIFY FORECAST TRACK AND INTENSITY 3 ISSUE IMPACT AND HAZARD INFORMATION 4 UPDATE AS UNCERTAINTY CHANGES

HAZARD / RISK / RESILIENCE CONCEPT

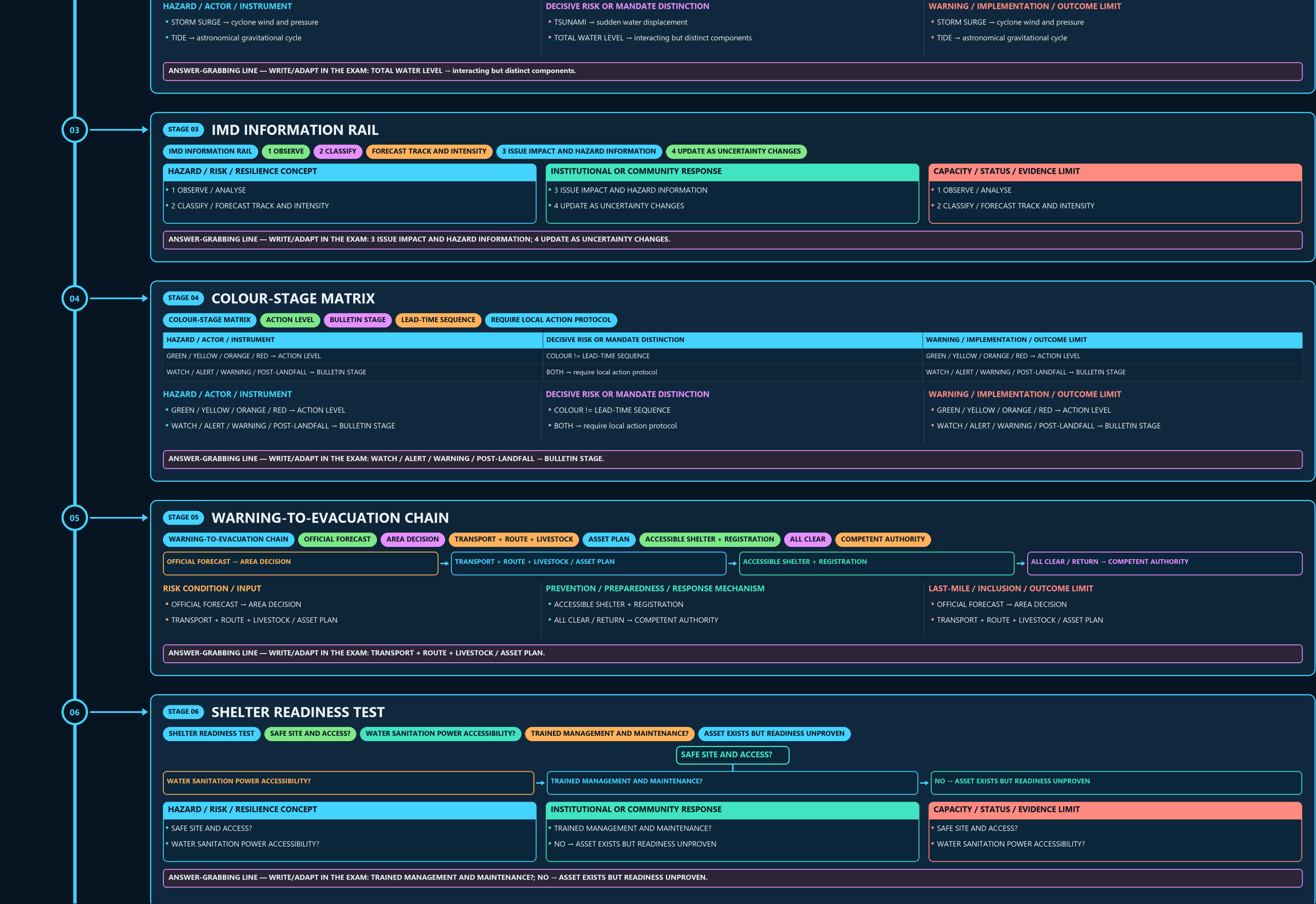
- 1 OBSERVE / ANALYSE
- 2 CLASSIFY / FORECAST TRACK AND INTENSITY

INSTITUTIONAL OR COMMUNITY RESPONSE

- 3 ISSUE IMPACT AND HAZARD INFORMATION
- 4 UPDATE AS UNCERTAINTY CHANGES

CAPACITY / STATUS / EVIDENCE LIMIT

- 1 OBSERVE / ANALYSE
- 2 CLASSIFY / FORECAST TRACK AND INTENSITY



06

STAGE 06 SHELTER READINESS TEST

SHELTER READINESS TEST SAFE SITE AND ACCESS? WATER SANITATION POWER ACCESSIBILITY? TRAINED MANAGEMENT AND MAINTENANCE? ASSET EXISTS BUT READINESS UNPROVEN

SAFE SITE AND ACCESS?

WATER SANITATION POWER ACCESSIBILITY?

TRAINED MANAGEMENT AND MAINTENANCE?

NO → ASSET EXISTS BUT READINESS UNPROVEN

HAZARD / RISK / RESILIENCE CONCEPT

- SAFE SITE AND ACCESS?
- WATER SANITATION POWER ACCESSIBILITY?

INSTITUTIONAL OR COMMUNITY RESPONSE

- TRAINED MANAGEMENT AND MAINTENANCE?
- NO → ASSET EXISTS BUT READINESS UNPROVEN

CAPACITY / STATUS / EVIDENCE LIMIT

- SAFE SITE AND ACCESS?
- WATER SANITATION POWER ACCESSIBILITY?

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAINED MANAGEMENT AND MAINTENANCE?; NO → ASSET EXISTS BUT READINESS UNPROVEN.

07

STAGE 07 CRITICAL-SERVICE WEB

CRITICAL-SERVICE WEB SUPPLY CHAIN EOC / HOSPITAL / RESCUE POWER / WATER / TELECOM

EOC / HOSPITAL / RESCUE

POWER / WATER / TELECOM

ROAD / PORT / SUPPLY CHAIN

REDUNDANCY → ASSESS → RESTORE

HAZARD / RISK / RESILIENCE CONCEPT

- EOC / HOSPITAL / RESCUE
- POWER / WATER / TELECOM

INSTITUTIONAL OR COMMUNITY RESPONSE

- ROAD / PORT / SUPPLY CHAIN
- REDUNDANCY → ASSESS → RESTORE

CAPACITY / STATUS / EVIDENCE LIMIT

- EOC / HOSPITAL / RESCUE
- POWER / WATER / TELECOM

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ROAD / PORT / SUPPLY CHAIN; REDUNDANCY → ASSESS → RESTORE.

08

STAGE 08 HOUSING-TO-ECOSYSTEM PORTFOLIO

HOUSING-TO-ECOSYSTEM PORTFOLIO SITING CODE COMPLIANCE MAINTENANCE LIFELINE HARDENING AND REDUNDANCY MANGROVE DUNE WETLAND PROTECTION LAYERED DEFENCE WITH RESIDUAL RISK

HAZARD / ACTOR / INSTRUMENT

HOUSING → siting code compliance maintenance
INFRASTRUCTURE → lifeline hardening and redundancy

DECISIVE RISK OR MANDATE DISTINCTION

ECOSYSTEM → mangrove dune wetland protection
RULE → layered defence with residual risk

WARNING / IMPLEMENTATION / OUTCOME LIMIT

HOUSING → siting code compliance maintenance
INFRASTRUCTURE → lifeline hardening and redundancy

HAZARD / ACTOR / INSTRUMENT

- HOUSING → siting code compliance maintenance
- INFRASTRUCTURE → lifeline hardening and redundancy

DECISIVE RISK OR MANDATE DISTINCTION

- ECOSYSTEM → mangrove dune wetland protection
- RULE → layered defence with residual risk

WARNING / IMPLEMENTATION / OUTCOME LIMIT

- HOUSING → siting code compliance maintenance
- INFRASTRUCTURE → lifeline hardening and redundancy

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ECOSYSTEM → mangrove dune wetland protection; RULE → layered defence with residual risk.

09

STAGE 09 NCRMP STATUS LADDER

NCRMP STATUS LADDER PROJECT APPROVAL ASSET CREATION PHASE COMPLETION LOCAL HAND-OFF OPERATION + MAINTENANCE + DRILLS CURRENT READINESS NEEDS SEPARATE EVIDENCE

PROJECT APPROVAL → ASSET CREATION

PHASE COMPLETION → STATE / LOCAL HAND-OFF

OPERATION + MAINTENANCE + DRILLS

CURRENT READINESS → needs separate evidence

RISK CONDITION / INPUT

- PROJECT APPROVAL → ASSET CREATION
- PHASE COMPLETION → STATE / LOCAL HAND-OFF

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM

- OPERATION + MAINTENANCE + DRILLS
- CURRENT READINESS → needs separate evidence

LAST-MILE / INCLUSION / OUTCOME LIMIT

- PROJECT APPROVAL → ASSET CREATION
- PHASE COMPLETION → STATE / LOCAL HAND-OFF

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OPERATION + MAINTENANCE + DRILLS; CURRENT READINESS → needs separate evidence.

09

STAGE 09 NCRMP STATUS LADDER

NCRMP STATUS LADDER PROJECT APPROVAL ASSET CREATION PHASE COMPLETION LOCAL HAND-OFF OPERATION + MAINTENANCE + DRILLS CURRENT READINESS NEEDS SEPARATE EVIDENCE

PROJECT APPROVAL → ASSET CREATION

PHASE COMPLETION → STATE / LOCAL HAND-OFF

OPERATION + MAINTENANCE + DRILLS

CURRENT READINESS → needs separate evidence

RISK CONDITION / INPUT

- PROJECT APPROVAL → ASSET CREATION
- PHASE COMPLETION → STATE / LOCAL HAND-OFF

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM

- OPERATION + MAINTENANCE + DRILLS
- CURRENT READINESS → needs separate evidence

LAST-MILE / INCLUSION / OUTCOME LIMIT

- PROJECT APPROVAL → ASSET CREATION
- PHASE COMPLETION → STATE / LOCAL HAND-OFF

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OPERATION + MAINTENANCE + DRILLS; CURRENT READINESS → needs separate evidence.

10

STAGE 10 LIVELIHOOD RECOVERY LOOP

LIVELIHOOD RECOVERY LOOP BOAT GEAR LIVESTOCK AND MARKET ACTION RELIEF + SOCIAL PROTECTION SAFER CONDITIONS BUILD BACK BETTER REDUCE FUTURE EXPOSURE

WARNING → BOAT GEAR LIVESTOCK AND MARKET ACTION

IMPACT → RELIEF + SOCIAL PROTECTION

REPAIR / REOPEN → SAFER CONDITIONS

BUILD BACK BETTER → REDUCE FUTURE EXPOSURE

HAZARD / RISK / RESILIENCE CONCEPT

- WARNING → BOAT GEAR LIVESTOCK AND MARKET ACTION
- IMPACT → RELIEF + SOCIAL PROTECTION

INSTITUTIONAL OR COMMUNITY RESPONSE

- REPAIR / REOPEN → SAFER CONDITIONS
- BUILD BACK BETTER → REDUCE FUTURE EXPOSURE

CAPACITY / STATUS / EVIDENCE LIMIT

- WARNING → BOAT GEAR LIVESTOCK AND MARKET ACTION
- IMPACT → RELIEF + SOCIAL PROTECTION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WARNING → BOAT GEAR LIVESTOCK AND MARKET ACTION.

11

STAGE 11 CYCLONE ANSWER SPINE

CYCLONE ANSWER SPINE DEFINE COMPOUND HAZARD SEPARATE SURGE TIDE TSUNAMI MAP IMD CLASSIFICATION FORECAST COLOUR AND BULLETIN STAGE CONNECT EVACUATION SHELTER LIFELINES HOUSING AND ECOSYSTEMS ADD LIVELIHOODS BBB MAINTENANCE AND OUTCOME QUALIFICATION

DEFINE COMPOUND HAZARD → SEPARATE SURGE TIDE TSUNAMI

MAP IMD CLASSIFICATION FORECAST COLOUR AND BULLETIN STAGE

CONNECT EVACUATION SHELTER LIFELINES HOUSING AND ECOSYSTEMS

ADD LIVELIHOODS BBB MAINTENANCE AND OUTCOME QUALIFICATION

HAZARD / RISK / RESILIENCE CONCEPT

- DEFINE COMPOUND HAZARD → SEPARATE SURGE TIDE TSUNAMI
- MAP IMD CLASSIFICATION FORECAST COLOUR AND BULLETIN STAGE

INSTITUTIONAL OR COMMUNITY RESPONSE

- CONNECT EVACUATION SHELTER LIFELINES HOUSING AND ECOSYSTEMS
- ADD LIVELIHOODS BBB MAINTENANCE AND OUTCOME QUALIFICATION

CAPACITY / STATUS / EVIDENCE LIMIT

- DEFINE COMPOUND HAZARD → SEPARATE SURGE TIDE TSUNAMI
- MAP IMD CLASSIFICATION FORECAST COLOUR AND BULLETIN STAGE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MAP IMD CLASSIFICATION FORECAST COLOUR AND BULLETIN STAGE.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT MULTI-SECTORAL PREMISE, EXPLICITLY STATED NDMA'S 2008 CYCLONE NCRMP'S INSTITUTIONAL STRUCTURE NDMA (UNDER MHA) IMPLEMENTS IN ASYMMETRIC COASTAL EXPOSURE, PRECISELY QUANTIFIED IN THE SOURCE

1994 ODISHA CYCLONE AS A CASCADING-HAZARD CASE AIRCRAFT PROBING OF CYCLONE (APC) AS AN UNDER-INSTITUTIONALISED

OPTIONAL RISK / RESILIENCE DEPTH

- Multi-sectoral premise, explicitly stated : NDMA's 2008 cyclone
- NCRMP's institutional structure : NDMA (under MHA) implements in
- Asymmetric coastal exposure, precisely quantified in the source :
- CAUTION: 1994 Odisha cyclone as a cascading-hazard case : cyclone

MANDATE, WARNING OR OUTCOME LIMIT

- CAUTION: Aircraft Probing of Cyclone (APC) as an under-institutionalised
- CAUTION: VisionIAS names "a lack of a fully automated and state-of-the-art
- CAUTION: VisionIAS's "lack of grass root level participation in disaster
- CAUTION: Category-based investment prioritisation is itself a trade-off.

COMPARATIVE HAZARD / GOVERNANCE NUANCE

- CAUTION: Saline-embankment/bio-shield maintenance versus new-construction
- NOT: The "NCRMP exists, therefore coastal States are protected"
- CAUTION: Network fragmentation across agencies : the explicit call for
- NCRMP's blended financing (NDMA/MHA-led, State-coordinated,

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.